

Data Flow Diagram (DFD)

Data Flow Diagram

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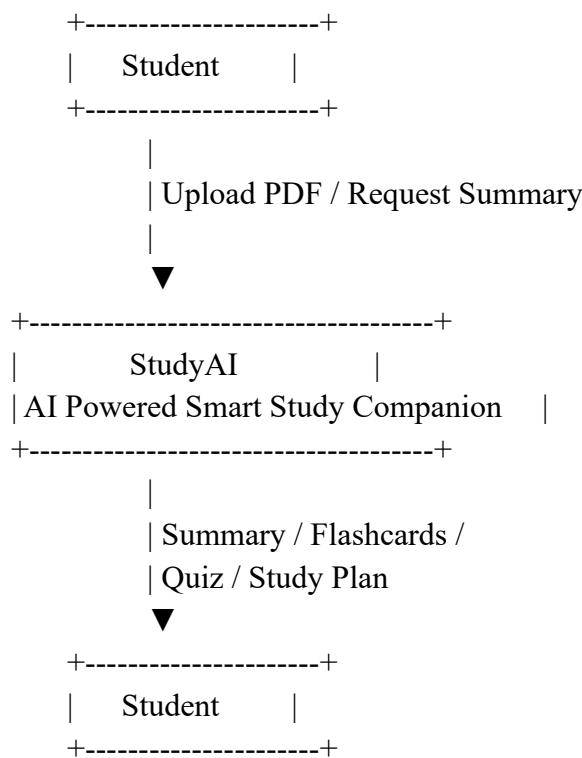
Team ID *****

Project Name StudyAI – AI Powered Smart Study Companion

Introduction

The Data Flow Diagram (DFD) illustrates how information moves through the StudyAI system. It shows the interaction between users, the web application, backend services, AI engine, and generated learning resources.

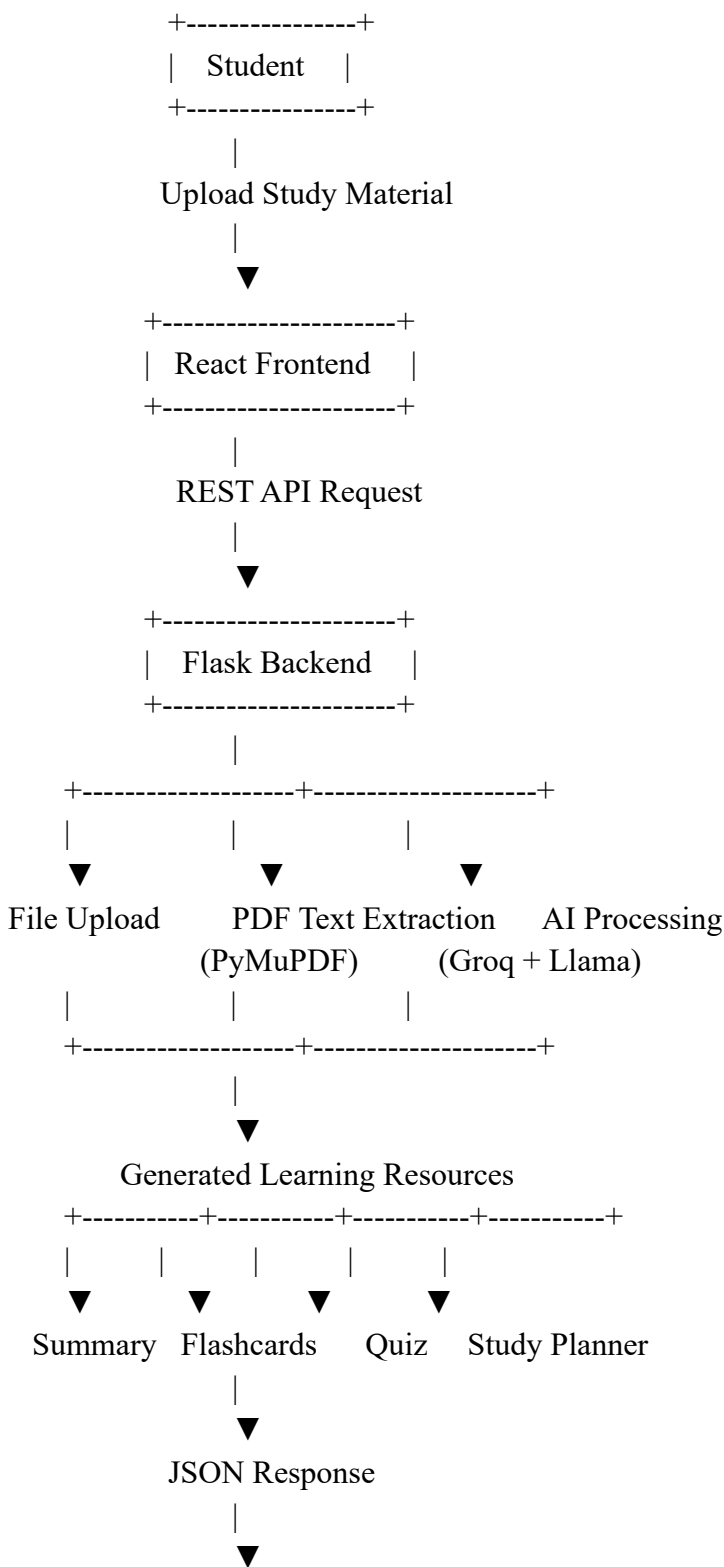
Level 0 DFD (Context Diagram)



Explanation

At the highest level, the student interacts with the StudyAI system by uploading study materials and requesting AI-powered learning resources. The system processes the input and returns summaries, flashcards, quizzes, and study plans.

Level 1 DFD



React Frontend



Student

Data Stores

Data Store	Purpose
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Uploads Folder	Stores uploaded study materials
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Local Storage	Stores uploaded filename for future processing
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Groq AI Model	Generates AI-based educational content
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Data Processes

Process	Description
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File Upload	Receives and validates uploaded documents
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Text Extraction	Extracts readable content from PDF using PyMuPDF
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AI Processing	Sends extracted text to Groq Llama model
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Summary Generation	Produces concise summaries
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Flashcard Generation	Creates revision flashcards (<i>planned</i>)
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Quiz Generation	Creates MCQs (<i>planned</i>)
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Study Planner	Generates personalized schedules (<i>planned</i>)
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Advantages of the DFD

- Clearly represents system workflow.
- Shows interaction between frontend, backend, and AI.
- Easy to understand and maintain.
- Supports future expansion of AI modules.